Day 27 Student Handout

Geometry Summer Credit | Required Capstone Modeling Work Session

Name:	Date:	Period:

Learning Targets

I can choose a realistic geometry modeling question. I can make assumptions, use formulas, and explain whether my answer is reasonable. I can prepare a short presentation using a rubric.

Lesson 27A: Capstone Launch and Work Session

Capstone Choices

A: Design a container/package with a capacity or material constraint. B: Model a real location using area, perimeter, and scale. C: Use right-triangle trig to estimate a height or distance. D: Propose your own model with teacher approval.

Warm-Up: Choose Your Model

1.	Which capstone option are you leaning toward? What real object, place, or situation will you model?

2.	What measurement will matter most: length, area, surface area, volume, scale, density, or trig? Explain.

3.	What is one assumption you may need to make so the problem is solvable?

Day 27 Student Handout

Continued | Modeling Plan

Modeling Plan

Problem	Work / notes
Question	What am I trying to decide, design, estimate, or compare?
Assumptions	What will I simplify? What will I ignore? What units will I use?
Geometry tools	Which formulas or ideas will I use: area, volume, scale, circle equation, trig, density, or coordinate geometry?

Day 27 Student Handout

Lesson 27B | Presentation and Reflection

Presentation Requirements

What to Include

Your presentation should include: the modeling question, a labeled diagram or sketch, assumptions, formulas, calculations, final decision, and one limitation or improvement.

4.	Draw or describe the diagram/model you will show in your presentation.

5.	Write your main formula setup. Include units.

6.	What final decision, estimate, or recommendation will your project make?

7.	What is one limitation of your model?